

SHAIK ABDUL RAZAK

🎓 B.Tech Computer Science & Engineering

📍 Kurnool, Andhra Pradesh, India

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Professional Summary

B.Tech Computer Science student and aspiring Machine Learning Engineer skilled in Python, DSA (C++), Java (Advanced), MySQL, Frontend Development, and Data Analytics. Published 5 research papers in healthcare AI with 3 accepted in reputed journals. Passionate about building innovative AI solutions and translating research into real-world applications. Seeking roles in Machine Learning Engineering and Software Development.

Technical Skills

Languages:	Python, Java, C/C++, SQL, JavaScript, HTML/CSS
ML/AI:	TensorFlow, PyTorch, Scikit-learn, NLP, Computer Vision, Ensemble Models, Deep Learning
Web Dev:	React, Django, Node.js, REST APIs, Flask, Tailwind CSS
Data Engineering:	Pandas, NumPy, Data Cleaning, Feature Engineering, Schema Harmonization, Deduplication
Tools & Platforms:	Git, Docker, AWS, Tableau, MATLAB, Google Colab, Excel, CI/CD, Linux
Databases:	MySQL, MongoDB, PostgreSQL, Firebase
Core CS:	Data Structures & Algorithms, OOP, Graph Algorithms, Dynamic Programming, System Design, DBMS, OS, Networks

Professional Experience

InternPro — Python Developer (Remote) Aug 2024 – Mar 2025

- Developed **5+ microservices** using Django, improving API response time by **30%**
- Built RESTful APIs handling **10,000+ daily requests** for enterprise client dashboards
- Automated ETL pipelines processing **50GB+ datasets**, saving **15+ hours/week** manual effort
- Implemented data validation and monitoring systems ensuring **99.9% data accuracy**

Forge Virtual Internships 2023 – 2024

- **Tata Group — GenAI Data Analytics:** Built analytics dashboards using Generative AI
- **Deloitte Australia — Data Analytics & Tech:** Solved business cases with data-driven solutions
- **Deloitte Australia — Technology Consulting:** Implemented cloud solutions and technology strategies
- **Deloitte Australia — Cyber Security:** Implemented security protocols and risk assessment frameworks
- **JPMorgan Chase — Software Engineering:** Developed financial data processing applications
- **AWS — Solutions Architecture:** Designed scalable cloud infrastructure solutions

Research & Technical Projects

MEGA-HEART — Comprehensive Cardiovascular Dataset through Multi-Source Clinical Data Integration

- Integrated **71,220+ patient records** from 3 heterogeneous cardiovascular datasets

- Developed schema harmonization pipeline reducing data inconsistencies by **40%**
- Implemented deduplication algorithm achieving **95% accuracy** in record matching
- Models trained on integrated data showed **15% better performance** than individual datasets

Comprehensive Diabetes Prediction Framework using Ensemble ML on Multi-Source Healthcare Data

- Merged **6 healthcare datasets** containing **580,000+ patient records**
- Handled **14.5 million missing entries** through advanced preprocessing pipeline
- Achieved **92.61% accuracy, 86.02% precision, 96.14% recall** using Random Forest ensemble
- Identified BMI, age, income level as most influential predictors

SmartRecruit AI — Intelligent Framework for Automated Talent Acquisition using Generative AI

- Combined n8n workflow automation with Google Gemini for intelligent candidate analysis
- Cut initial screening time by **80%** and improved matching accuracy by **31%**
- Boosted candidate response rates by **60%** through personalized communications
- Integrated Firebase for real-time data handling and Next.js for frontend interface

DR-MERGE — Large-Scale Multi-Source Retinal Image Dataset for Diabetic Retinopathy

- Consolidated **17,976 unique retinal images** from 4 public datasets
- Developed systematic integration pipeline with MD5-based deduplication
- Removed **128 duplicate images** ensuring dataset quality and uniqueness
- Created one of the largest unified retinal image collections for DR research

BraTumDB-1.0 — Unified Brain Tumor MRI Dataset Benchmark

- Merged **6 public sources** into unified dataset of **8,927 unique brain MRI scans**
- Removed **5,727 duplicate images (39.1%)** using MD5 hashing techniques
- Standardized labeling and processed all images to uniform **256×256 pixel size**
- Created ready-to-use train-validation-test splits (70-15-15 ratio) with detailed metadata

Temporal Retinal DNA (t-DNA) — DR Progression Prediction Framework

- Developed LSTM-based framework predicting 3-year DR progression with **89% accuracy**
- Incorporated temporal features from sequential retinal scans over multiple visits
- Published in **Journal of Medical AI** (Impact Factor: 6.5)
- Enabled early intervention through accurate progression forecasting

Real-Time Diabetic Retinopathy Detection System

- Built CNN-based system detecting DR stages with **94% accuracy** from retinal images
- Reduced screening time by **70%** through automated analysis
- Deployed as web application serving **1000+ monthly screenings**
- Integrated with electronic health record systems for seamless clinical workflow

Speech Emotion Recognition System

- Built real-time emotion recognition model classifying **7 emotions** from speech
- Achieved **87% accuracy** using MFCC features and SVM/MLP classifiers
- Implemented noise reduction and feature extraction pipeline
- *Tech:* Librosa, Scikit-learn, Signal Processing, Flask

Fake News Detection System

- Developed NLP-based classifier detecting fake news with **92% accuracy**
- Processed **10,000+ news articles** using BERT embeddings and attention mechanisms
- Implemented real-time verification against trusted sources
- *Tech:* Transformers, BERT, Flask, MongoDB, Web Scraping

Customer Churn Prediction System

- Implemented ensemble models predicting customer churn with **88% precision**
- Identified key churn factors improving retention strategies by **25%**
- Developed customer segmentation using clustering algorithms
- *Tech:* XGBoost, Random Forest, Feature Engineering, Tableau Dashboard

Face Recognition-Based Attendance System

- Developed automated attendance system using face recognition technology
- Achieved **96% recognition accuracy** under varying lighting conditions
- Integrated with college management system for automatic reporting
- *Tech:* OpenCV, Face Recognition Libraries, Python, SQLite

Research Publications

- K. S. Chunchu, S. A. Razak, S. Saniya, A. K. Morya, and R. P. Kumar, “DR-MERGE: A Large-Scale Multi-Source Retinal Image Dataset for Diabetic Retinopathy via Systematic Integration and Deduplication”, IJASIS, pp. 777–792, Jan. 2026, doi: 10.29284/mfph5424.
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- Comprehensive Diabetes Prediction Framework Using Ensemble ML on Multi-Source Healthcare Data – Under review
- SmartRecruit AI: Intelligent Framework for Automated Talent Acquisition Using Generative AI – Under review

Conferences

- MEGA-HEART: Comprehensive Cardiovascular Dataset through Multi-Source Clinical Data Integration – ICCIC 12-13 DECEMBER 2025

Internships

InternPro — Machine Learning Intern

Jul – Aug 2025

- Completed **6-week intensive ML internship** under mentorship of Donesh Kumbhare
- Developed end-to-end ML pipelines for customer segmentation and recommendation systems
- Implemented advanced feature engineering improving model accuracy by **12%**
- [View Certificate]

StaxTech — Front-End Developer Intern

Jul 2025

- Built **10+ responsive UI components** using React and Tailwind CSS
- Improved website performance score from **65 to 92** on Google Lighthouse
- [View Certificate - STAXP617]

GWING — Data Analyst Intern

Sep – Oct 2025

- Performed data analysis and visualization for business intelligence insights
- Created interactive dashboards using Python and Excel for decision-making
- [View Certificate - GWING817563]

- Solved **150+ algorithmic problems** with focus on optimization
- Implemented advanced data structures (Segment Trees, Trie, Union-Find)
- [View Certificate]

VaultofCodes — Java Programming Intern

Nov 2023 – Jan 2024

- Developed **3 Java applications** implementing OOP principles
- Built backend logic for inventory management system
- [View Certificate]

Certifications

Forage Virtual Internships

- **Tata Group — GenAI Data Analytics** [View Certificate]
- **Deloitte Australia — Data Analytics & Technology** [View Certificate]
- **Deloitte Australia — Technology Consulting** [View Certificate]
- **Accenture — Software Engineering** [View Certificate]
- **AWS — Solutions Architecture** [View Certificate]

IBM SkillsBuild Certifications

- **Java Fundamentals — IBM SkillsBuild** [View Certificate]
- **Python Programming — IBM SkillsBuild** [View Certificate]

Technical Internship Certificates

- **StaxTech — Front-End Development (Jul 2025)** [View Certificate - STAXP617]
- **GWING Software Technologies — Data Analyst Internship (Sep-Oct 2025)** [View Certificate - GWING817563]
- **VaultofCodes — Java Programming Internship** [View Certificate]
- **InternPro — Machine Learning Internship (Jul-Aug 2025)** [View Certificate]
- **SystemTron — DSA in C++ Internship** [View Certificate]

HP LIFE Certifications

- **AI for Beginners**
- **Data Science & Analytics**

Additional Professional Training

- **Advanced Software Engineering — Forage**
- **Deloitte Data Analytics — Forage**
- **Deloitte Technology — Forage**

Achievements

- **Automation & Generative AI – 24 Hours Hackathon 2025, KLH University**
 - Built automated document processing system reducing manual work by **75%**
 - Integrated OCR and LLM for intelligent document analysis
- **Top 20 Teams — PSCMR University Hackathon (600+ teams)**
 - Developed mental health chatbot using NLP and sentiment analysis
 - Provided personalized coping strategies based on user conversations
- Solved **100+ DSA problems** on LeetCode and CodeChef (**3-star rating**)
- Published **2 research papers** in reputed journals and conferences
- Maintained **9.90/10 CGPA** (Top 2% of department)

Education

Bachelor of Technology — Computer Science and Engineering

Chaitanya Deemed to be University, Hyderabad

2023 – 2027

CGPA: 9.90/10 (Department Rank: Top 2 percent)

Intermediate (MPC) — Aaditya Junior College, Kurnool

State Board of Andhra Pradesh

2021 – 2023

Percentage: 92 percent

Secondary Education (Class X) — Government High School

State Board

2020 – 2021

Percentage: 85 percent

Additional Information

Languages:	English (Fluent), Telugu (Native), Hindi (Fluent), Urdu (Conversational)
Technical Profiles:	Google Scholar — Kaggle — CodePen — GitLab — Medium
Technical Writing:	Published 2+ technical articles on Medium covering AI/ML, Web Development, and Research
Open Source:	Active contributor on GitHub with 48+ repositories
Competitive Prog:	Regular participant in CodeChef, LeetCode contests
Interests:	Volleyball, Hiking, Technical Blogging, Open Source Projects
Availability:	Immediately available for full-time/internship roles